ShareSphere — Hackathon Submission Deliverable Report

Problem Statement: “Unlock Underused Equipment and Spaces”

Project: ShareSphere (Trusted Resource Sharing & Accountability Platform)

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1. Problem Statement & Target Users

Problem

Universities, colleges, laboratories, makerspaces, community organizations, and small tech firms own valuable equipment and spaces that often remain dormant 70% of the working week. At the same time, students, graduate researchers, creators, and early-stage startup teams lack affordable access to high-grade infrastructure. Traditional shared calendar systems (like Google Calendar or basic room bookers) fail because:

1. They lack **safety gate enforcement**: anyone can book an industrial laser cutter or cleanroom microscope without safety certifications.
2. They offer **no condition accountability**: if a $600 lens filter is scratched or cables go missing, there is no proof of who caused the damage.
3. They allow **resource hoarding**: without fair-use limits, a few individuals monopolize scarce tools.
4. They lack **explainable discovery**: users often do not know which equipment fits their experimental requirements.

Target Users

- **Borrowers**: Students, graduate researchers, student startup founders, and cross-departmental creators.
- **Resource Owners**: Principal investigators, lab technicians, makerspace directors, and equipment custodians.
- **Institution Managers**: Department chairs, facilities deans, safety officers, and dispute arbiters.
- **Administrators**: Campus IT and campus-wide governance officers.

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2. Solution & Core Workflows

ShareSphere provides an end-to-end trusted resource-sharing lifecycle:

1. **Safety First**: Every resource listing defines required safety training credentials (e.g. `laser_safety_l2`, `cleanroom_iso6`). The backend verifies the user's institutional qualification records. Uncertified borrowers cannot book restricted tools.
2. **Explainable AI Matching**: Users describe their research goal in natural language (e.g., *"Need to debug high-frequency noise on custom FPGA PCB"*). The engine evaluates equipment capabilities, verifies safety certifications, checks live calendar availability, and returns top matches with bulleted explanations.
3. **Dedicated Handover Interface**: Check-in captures server timestamps, condition checklists, and notes. Checkout captures clean return condition, cables returned, and missing parts flags.

4. Append-Only Immutable Audit Trail: Every transaction is permanently committed to `audit_events`.

5. Fair Allocation & Waiting Lists: Automatic FIFO queue positioning with automatic notifications if prior reservations are cancelled.

6. Dispute & Incident Arbitration: When discrepancies occur, human managers review evidence and record decisions without arbitrary AI penalties.

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3. Technology Stack & Design Decisions

• **Frontend**: React 19, TypeScript, Vite, Tailwind CSS v4, Lucide React.

• *Rationale*: Instant responsiveness, typed safety, institutional SaaS aesthetic (deep navy typography, crisp slate canvas, restrained indigo accents).

• **Backend**: Python 3.14, FastAPI, Uvicorn, Pydantic v2.

• *Rationale*: High performance, automatic schema validation, auto-generated OpenAPI documentation at `/docs`, asynchronous speed.

• **Database**: SQLAlchemy 2.0 with SQLite development fallback and seamless PostgreSQL deployment capability via `DATABASE_URL`.

• **Security**: PBKDF2 salted password hashing, JWT Bearer tokens, server-side role dependency injection.

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4. Privacy, Security & Fairness

1. Security:

• Passwords hashed using PBKDF2 with SHA-256 and unique institutional salt.

• Roles strictly enforced on the backend via `require_roles(['owner', 'manager'])`.

• Server-generated timestamps prevent client clock tampering.

2. Privacy:

• Minimal PII collection (name, institutional email, department).

• Audit trail redacts passwords and secrets.

3. Fairness:

• Rolling 7-day quota prevents hoarding.

• Transparent FIFO waiting list with queue position visibility.

• No secret priority algorithms; all allocation rules are disclosed to the borrower.

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5. 3-Minute Judge Presentation Script

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[0:00 - 0:30] THE PROBLEM & CONSOLE OVERVIEW
"Good afternoon judges. Millions of dollars of lab equipment sits idle in universities because lab directors t

[0:30 - 1:00] SAFETY GATE ENFORCEMENT DEMO
"First, let's look at safety. I switch to Alex Chen—an undergraduate who only holds general safety. Alex opens
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[1:00 - 1:45] EXPLAINABLE AI MATCHING & LEGITIMATE BOOKING
"Now, let's switch to Dr. Maya Lin, a certified researcher. She opens the 'AI Equipment Matcher' and types: 'N

[1:45 - 2:30] OWNER APPROVAL & HANDOVER CHECK-IN
"Next, I switch to Prof. Marcus Vance, the Lab Director. In 'Approvals & Bookings', he reviews Maya's request

[2:30 - 3:00] RETURN, INCIDENTS & IMMUTABLE AUDIT TRAIL
"Upon return, condition is inspected. In our seeded scenario, on booking BK-7805 for the Sony FX6 Cinema kit,
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6. Known Limitations & Future Development

1. Physical Hardware Locks: Currently relies on digital signoff and staff handovers. Future work will integrate NFC/RFID badge readers and electronic smart-lockers.

2. Automated Calibration Telemetry: In the future, IoT instruments can directly report runtime telemetry (e.g., laser tube hours or SEM vacuum pressure) into the return record.

3. Third-Party LMS Integration: Can connect to university Canvas/Blackboard LMS to automatically sync completed lab safety certifications.

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7. AI Disclosure

• **AI Coding Assistance**: Google DeepMind Antigravity was used as an agentic pair-programming tool during the designated hackathon build window.

• **AI System Features**: Deterministic explainable keyword and capability matching engine built into the backend, with optional Google Generative AI enhancement for natural language query parsing.